

Multiple Choice

1. **Answer:** A

Options: A) Both species benefit equally; B) One species benefits, and the other is significantly harmed; C) One species harms the other without any benefit; D) Both species harm each other; E) The species exchange benefits, but one benefits more than the other

Note: In a commensalistic relationship, one species benefits while the other is neither helped nor harmed, so the correct characterization of this relationship is that both species do not benefit equally.

2. **Answer:** D

Options: A) Asexual reproduction; B) Coadapted gene complexes; C) Epistasis; D) Random mating

Note: Random mating does not promote linkage disequilibrium because it leads to the random assortment of alleles, which minimizes the correlation between alleles at different loci and promotes genetic diversity within a population.

3. **Answer:** C

Options: A) They carry blood towards the heart.; B) They always carry deoxygenated blood.; C) They carry blood away from the heart.; D) They contain valves that prevent backflow.; E) They always carry oxygenated blood.

Note: The correct answer is accurate because arteries are blood vessels specifically designed to transport oxygenated blood away from the heart to the body's tissues, playing a crucial role in the circulatory system.

4. **Answer:** A

Options: A) Taste, wind direction, humidity; B) Atmospheric oxygen levels, plant growth, tidal patterns; C) Infrared radiation, sound frequency, barometric pressure; D) Moon phases, echo location, gravitational pull; E) Sunlight intensity, water current, soil composition

Note: Animals utilize a combination of sensory cues, including taste to detect environmental changes, wind direction for navigation, and humidity to assess weather conditions, all of which are crucial for successful migration.

5. **Answer:** E

Options: A) Thick, lush foliage; B) Smooth, glossy leaves; C) High rates of transpiration; D) Shallow root systems; E) Spines instead of leaves

Note: Spines instead of leaves are an adaptation that reduces water loss by minimizing the surface area exposed to the sun and decreasing transpiration, which is crucial for survival in high-temperature environments.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Milk spoils primarily due to bacterial activity, which leads to fermentation and the production of off-flavors and curdling, rather than solely due to the separation of fat content at lower temperatures.

7. **Answer:** True

Options: A) True; B) False

Note: Endotherms, or warm-blooded animals, possess physiological mechanisms such as thermoregulation that allow them to maintain a constant internal body temperature and metabolic rate, regardless of external temperature fluctuations.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because if 46% of the loci are heterozygous, then the average homozygosity should be 54%, calculated as $1 - 0.46$.

9. **Answer:** True

Options: A) True; B) False

Note: The neritic zone, located above the continental shelf, is characterized by shallow waters that receive ample sunlight, promoting photosynthesis and supporting high nutrient availability often enhanced by upwelling currents.

10. **Answer:** True

Options: A) True; B) False

Note: Muscle tissue is classified as one of the four primary types of tissue in the body, distinct from connective tissue, because its primary function is to facilitate movement through contraction, whereas connective tissue primarily supports, binds, and protects other tissues and organs.

Long Form Response

11. **Answer:** Genetic crosses involving strains of highly inbred mice can help elucidate the genetic basis of transplant rejection or acceptance by allowing researchers to study the inheritance patterns of immune response genes. These inbred strains have well-defined genetic backgrounds, which makes it easier to identify specific alleles responsible for the recognition of foreign tissues. By analyzing the outcomes of transplants between different strains, researchers can correlate genetic markers with either successful acceptance or rejection of grafts. This approach provides insights into the major histocompatibility complex (MHC) genes and other immune-related loci that govern compatibility, helping to enhance our understanding of transplant immunology and improve transplant outcomes in clinical settings. Thus, the genetic basis of transplant rejection or acceptance is revealed through the systematic examination of these controlled genetic crosses.

Note: correct because highly inbred mice provide a controlled genetic environment that enables researchers to identify and link specific genetic variations to immune responses in transplant scenarios, thereby revealing the genetic mechanisms underlying transplant rejection or acceptance.

12. **Answer:** All living organisms share fundamental nutritional requirements, including the need for energy, carbon, and specific nutrients to sustain metabolic processes. However, these requirements manifest differently across various groups. Phototrophs, like plants, derive energy from sunlight and utilize carbon dioxide to produce organic compounds via photosynthesis, making them autotrophs. In contrast, chemotrophs obtain energy through chemical reactions, with some being autotrophs that fix inorganic carbon, while others are heterotrophs that rely on organic matter. Heterotrophs, including animals and certain fungi, must consume organic substances for both energy and carbon. Thus, while all organisms require basic nutrients, the methods of obtaining energy and carbon differ significantly among phototrophs, chemotrophs, autotrophs, and heterotrophs.

Note: correct because it identifies the essential nutritional requirements common to all life forms while clearly distinguishing between the different energy and carbon sourcing strategies of phototrophs, chemotrophs, autotrophs, and heterotrophs.